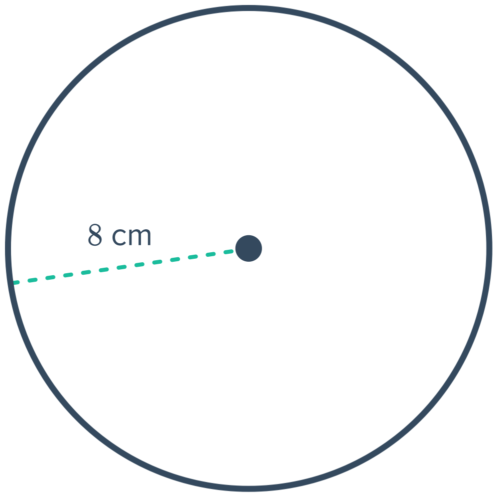
Review: Circumference and area

1 Find the circumference of the following circles, correct to one decimal place:

a



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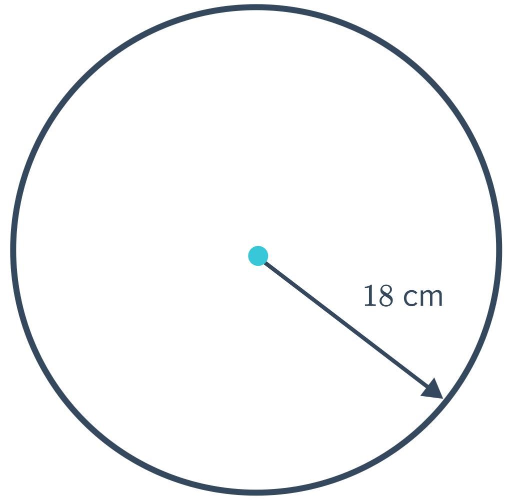
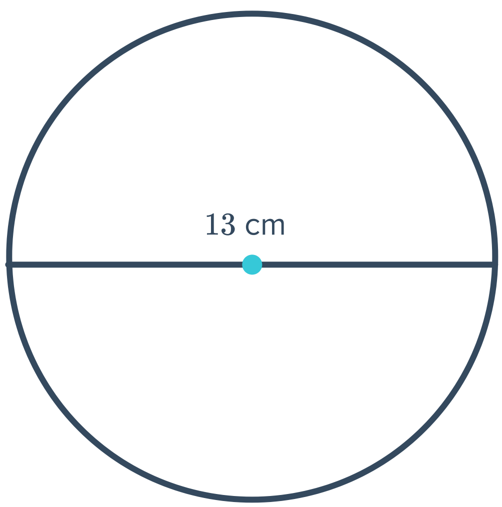


c



2 Find the circumference of the following circles, correct to one decimal place:

a A circle with a radius of 7 cm.

b A circle with a radius of 27 cm.

c A circle with a radius of 1.8 cm.

d A circle with a diameter of 38 cm.

3 Find the radius of the following circles, correct to one decimal place:

a A circle with circumference 18 m.

b A circle with circumference 14.9 m.

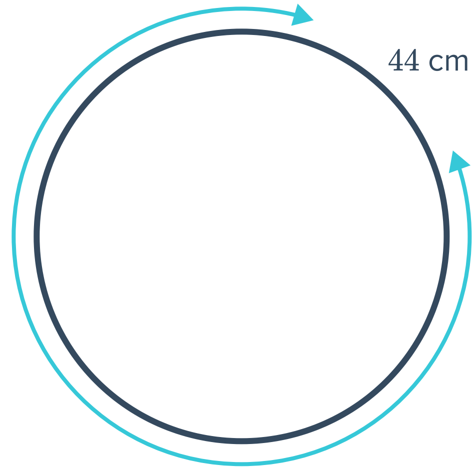
4 Find the diameter of the following circles, c[redacted]t to one decimal place:

a A circle with circumference 44 m.

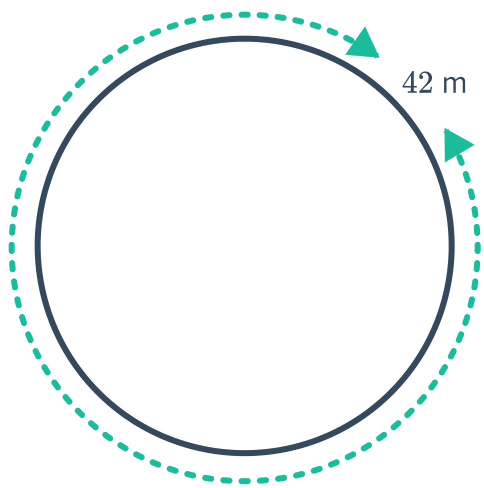
b A circle with circumference 70 m.

c A circle with circumference 24.5 cm.

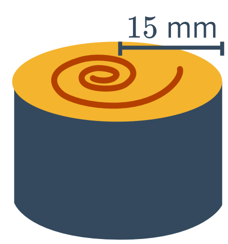
5 Find the radius of the circle shown with circumference of 44 cm. Round your answer to two decimal places.



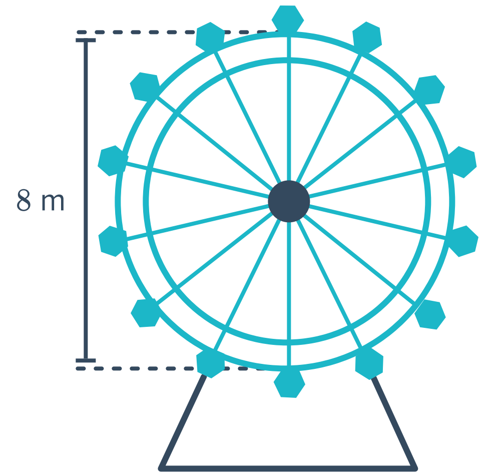
6 Find the diameter of the circle shown with circumference of 42 m. Round your answer to two decimal places.



7 Find the length of the strip of seaweed around the outside of the sushi. Give your answer correct to one decimal place.



8 Find the circumference of the Ferris wheel. Give your answer correct to one decimal place.



9 The bottom of a flower pot with uniform cross-section has a radius of $16\ \mathrm{cm}$. Find the circumference of the bottom of the flower pot. Round your answer to one decimal place.

10 A scooter tire has a diameter of $34\ \mathrm{cm}$. Find the circumference of the tire. Round your answer to one decimal place.

11 A coin has a diameter of $2.2\ \mathrm{cm}$. If the coin is rolled through 45 complete revolutions, find the distance it will travel. Give your answer correct to one decimal place.

12 A circular running track has a diameter of $23\ \mathrm{m}$. Find the number of laps must be completed to run $1600\ \mathrm{m}$. Give your answer correct to one decimal place.

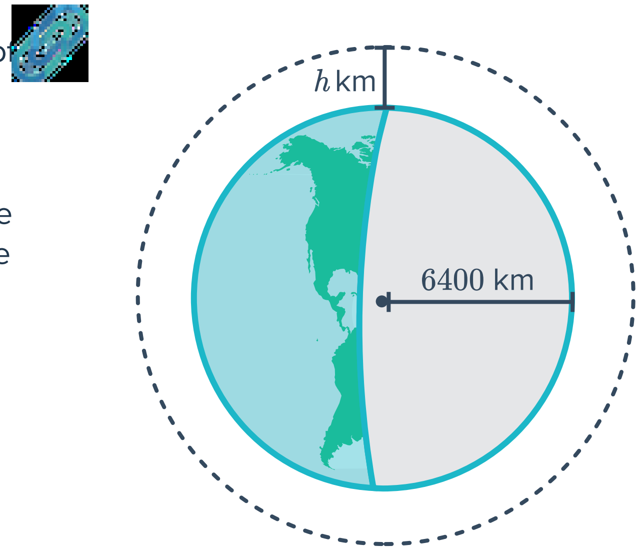
13 The wheel of Roxanne's bicycle has a radius of $22\ \mathrm{cm}$. Find the distance she would cycle if the wheels made 250 complete revolutions. Give your answer in meters, correct to one decimal place.

14 Carl is performing an experiment by spinning a metal weight around on the end of a nylon thread. Find the distance the metal weight travels if it completes 40 revolutions on the end of a $0.65\ \mathrm{m}$ thread. Give your answer correct to one decimal place.

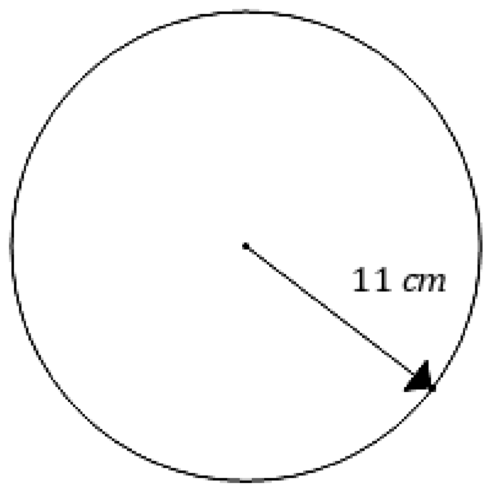
15 A shallow circular wading pool has a diameter of $4\ \mathrm{m}$. A deeper circular swimming pool has a radius of $16\ \mathrm{m}$. How many times greater is the circumference of the swimming pool than the wading pool?

16 The diameter of the front wheel of a car is $0.4\ \mathrm{m}$. How many complete revolutions will the wheel make if the car travels a distance of $50\ \mathrm{km}$?

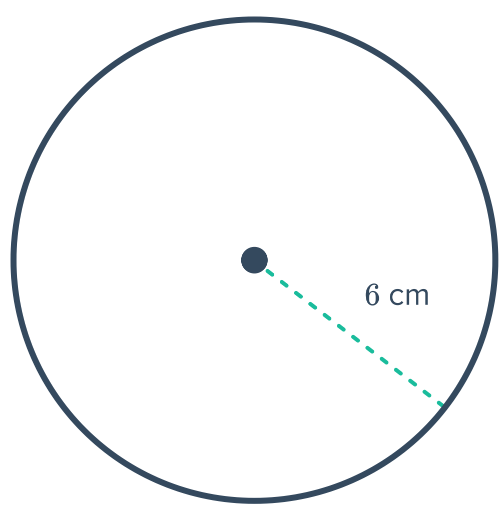
17 A satellite is orbiting the Earth at a height of h km above the Earth's surface. In one complete orbit, the satellite travels a distance of $41\,664$ km.
If the radius of the Earth is 6400 km, find the height of the satellite above the Earth. Give your answer correct to one decimal place.



18 Find the exact area of the circle shown:



19 Find the area of the circle shown, correct to one decimal place.



21 Find the area of the following circles, correct to one decimal place:

a A circle with a radius of 9 cm.

b A circle with a diameter of 24 cm.

22 The radius of a circular baking tray is 10 cm. Find its area, correct to two decimal places.

23 The area of a circle is 352 cm^2.

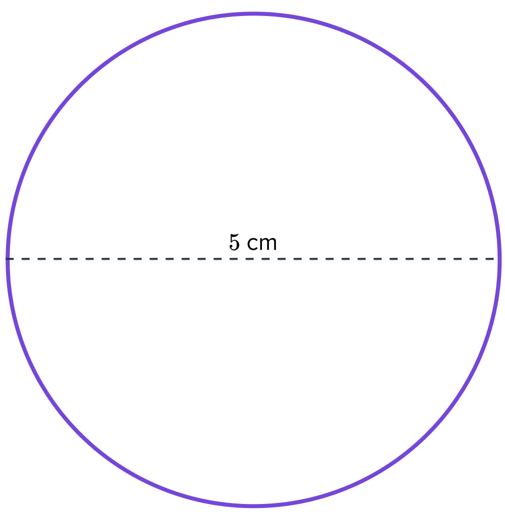
a If its radius is r cm, find r, correct to two decimal places.

b Hence, find the circumference of the circle. Round your answer to one decimal place.

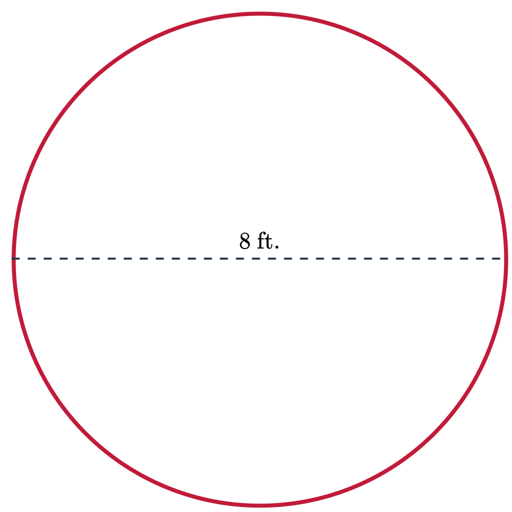
Additional questions

24 Find the circumference of each of the following circles:

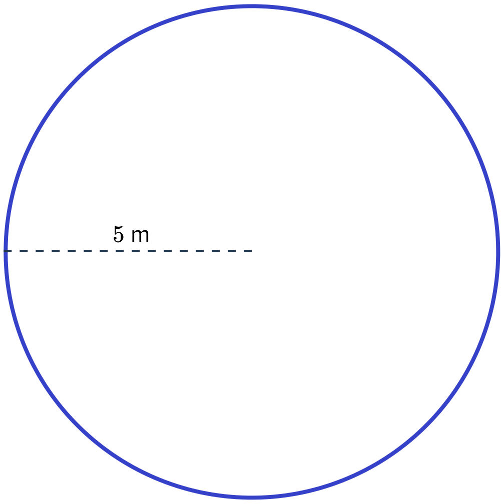
a

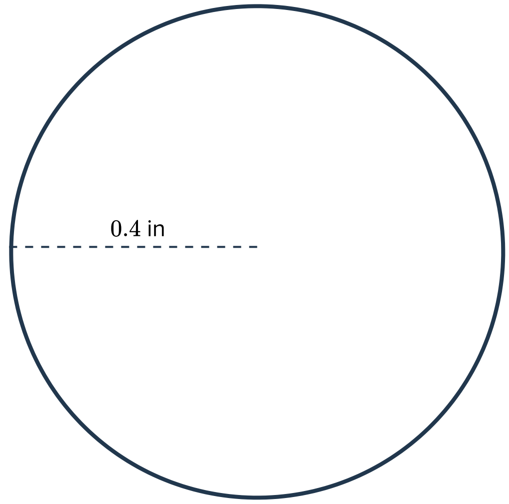


b



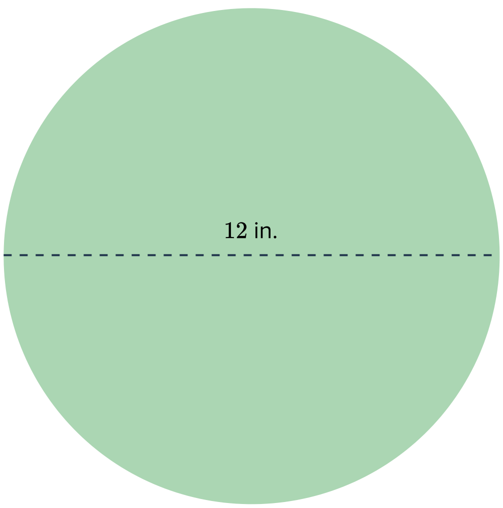
c





25 Find the area of each of the following circles:

a



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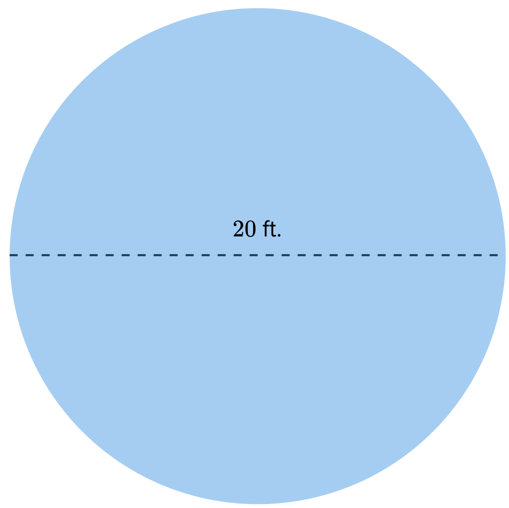

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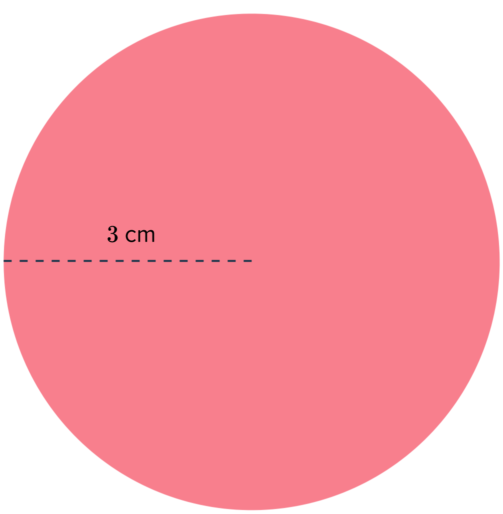
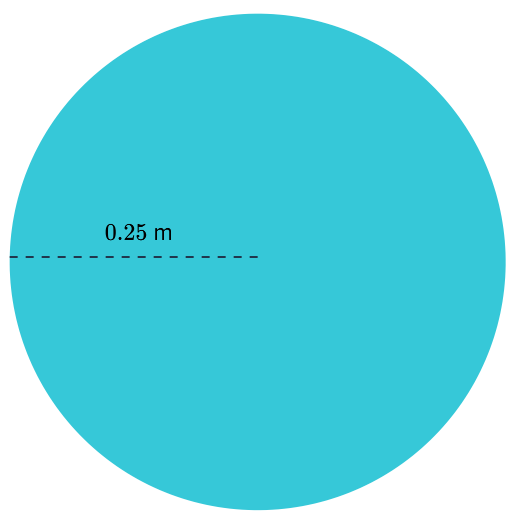


d



26 Find the area and circumference of each of [redacted]llowing circles:

a $r = 13 \text{ in}$

b $d = 3 \text{ yd}$

27 The area of a circle is 113 ft^2. Find the radius.

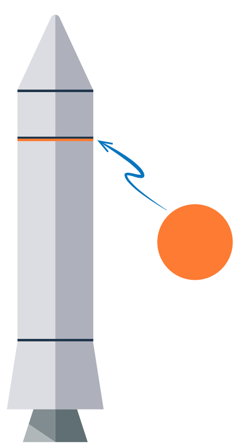
28 The radius of a circle is 14 in. Find the area.

29 The wheels on a children's bike have a diameter of 17 in.
About how far does each wheel travel on the ground in order to complete one full revolution?

30 A box of grass seed covers 350 ft^2 of ground. Maclay Gardens has a circular field that is 75 feet across.
How many boxes of grass seed are required to cover the field?

31 A circular track has a diameter of 200 m.
About how far is it around the track?

32 The engineering team at Rocket Surgery are building a rocket for an upcoming mission to Mars.
A critical piece is the circular connective disc that connects the booster rocket to the rest of the spacecraft. This disc must completely cover the top of the booster rocket.
The booster rocket has a circumference of precisely 713.5π centimeters.

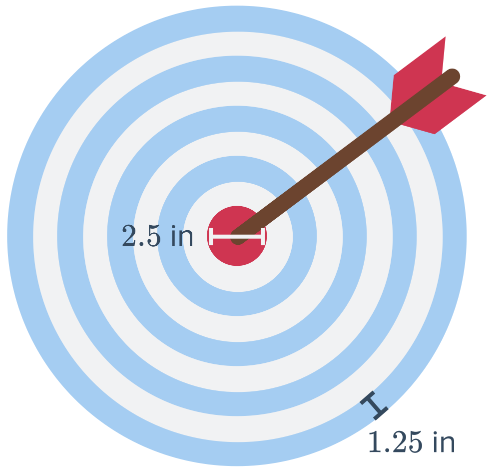


a Find the required area of the connective disc.

b An engineer at Rocket Surgery decided to use the approximation 3.14 instead of π.
Find the area of the connective disc using the engineer's approximation.

c If the connective disc is more than $100 \, \text{cm}^2$ too small, the disc will malfunction resulting in catastrophic launch rocket failure.
Is there a risk of malfunction if the disc is built according to the engineer's calculation?

33 Humberto has a custom dartboard, where each ring has a width of 1.25 in. The bullseye, or the center circle on the dartboard has a total width of 2.5 in.



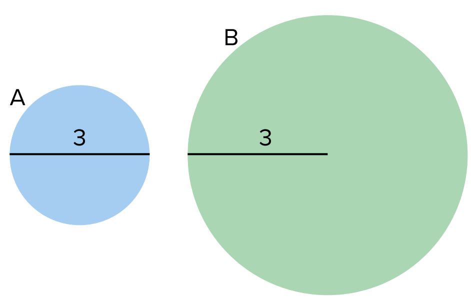
a What is the area of the entire dart board?

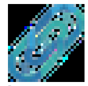
b What is the area of the dart board, excluding the center bullseye?

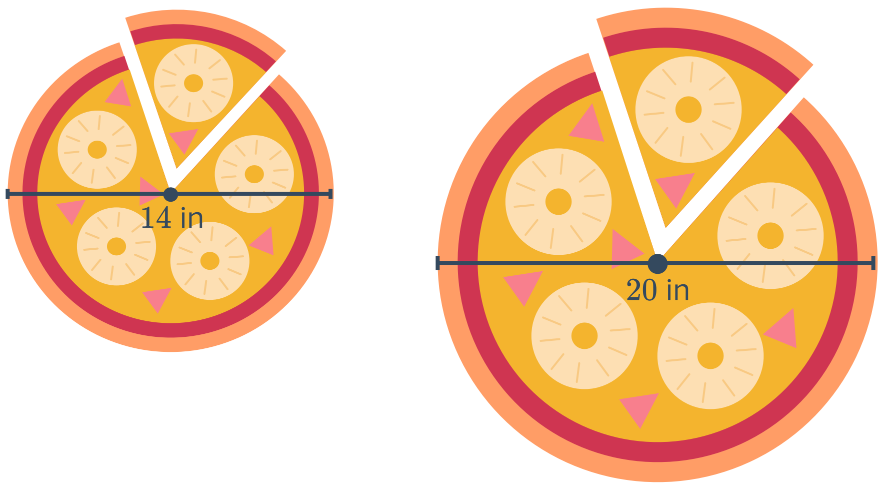
c What is the probability of not hitting the bullseye on the dartboard?

34 The garden at Hillsborough Community College has a circular koi pond at the center of campus. Jameela walked around the perimeter of the pond and using her stride length, she estimated the distance around the pond to be 100 feet.
Based on Jameela's estimation, will a 35-foot long bridge be long enough to to span the widest part of the pond? Explain.

35 Larry claims that the area of circle B is twice the area of circle A.
Is Larry correct? Explain how you know.

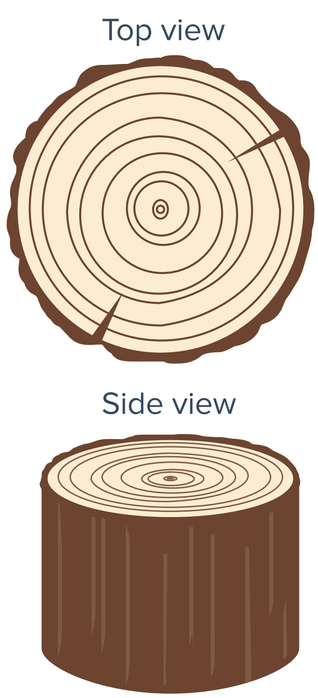


36 Faith and Carlena are in charge of ordering  for their friends. For $\$19.99$, they can either buy two 14-inch pizzas or one 20 -inch pizza.



Which option should Faith and Carlena buy in order to get the most pizza for their money?

37 The rings on a cross section of tree trunk can be used to estimate the age of a tree, where each ring can be counted as one year of the tree's life:



The average width of a tree ring for a particular grove of Slash Pine Trees is 0.75 in.

a Using only a tape measure, describe how you could determine the approximate age of a Slash Pine Tree without cutting it down.

b Use your method from part a) to determine the age of a Slash Pine Tree that has a circumference of 45in.